

PYTHON DATA ANALYSIS PORTFOLIO

Automobile Pricing Analytics

A Python-Based Empirical Study of Vehicle Characteristics, Price Relationships, and Regression Model Generalization

Portfolio Report

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Analytical workflow: Data Importing → Data Cleaning & Feature Engineering → Exploratory Data Analysis
→ Regression Model Development → Model Evaluation & Selection

*Dataset: UCI Machine Learning Repository Automobile dataset (1985 Ward's Automotive Yearbook)
Companion report to five reproducible Python notebooks*

Abstract

This portfolio project examines **how observable automobile characteristics are associated with vehicle price** and **how alternative regression specifications generalize beyond the estimation sample**. The analysis uses the **UCI Automobile dataset**, a multivariate dataset derived from the **1985 Ward's Automotive Yearbook**. The raw file contains 205 observations and 26 columns in the working representation used in the notebooks; four observations lack price and are excluded from target-based analysis, producing a cleaned analytical sample of 201 vehicles and 29 variables after feature preparation. The workflow is implemented in **Python** and organized into five reproducible stages: **data importing and initial assessment**, **data cleaning and feature engineering**, **exploratory data analysis (EDA)**, **regression model development**, and **regression model evaluation and selection**.

EDA identifies strong positive associations between **price** and **engine size** ($r = 0.872$), **curb weight** ($r = 0.834$), **horsepower** ($r = 0.810$), and **vehicle width** ($r = 0.751$), while highway MPG is negatively associated with price ($r = -0.705$). In model development, a **four-predictor multiple linear regression (MLR)** using horsepower, curb weight, engine size, and highway MPG achieves an in-sample R^2 of 0.809, compared with 0.761 for an **engine-size simple linear regression** and 0.497 for a **highway-MPG simple linear regression**. In the final evaluation, the MLR also leads both 5-fold cross-validation (mean $R^2 = 0.767$) and the untouched holdout test set ($R^2 = 0.764$; RMSE = \$5,377; MAE = \$3,769). A **polynomial highway-MPG** specification improves as complexity increases through **degree 6** in cross-validation, after which validation performance deteriorates sharply; nevertheless, the selected degree-6 polynomial performs materially worse than the MLR on the holdout sample ($R^2 = 0.546$).

The project demonstrates a **regression-centered business analytics workflow**. Its contribution is methodological: it connects data quality decisions, empirical exploration, model specification, generalization testing, and cautious interpretation in one coherent analysis. The results describe associations in a small historical dataset and should not be interpreted as causal estimates or a contemporary automobile valuation model.

Portfolio Contribution

For a business/data analytics portfolio, this project demonstrates the ability to convert a raw external dataset into a reproducible analytical study using Python; justify cleaning decisions; distinguish descriptive association from causal inference; translate EDA findings into regression specifications; evaluate model complexity and overfitting; and communicate quantitative evidence in a form suitable for managerial and research audiences.

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1. Introduction and Analytical Motivation

Pricing is a central analytical problem in business because observed prices often reflect multiple product attributes simultaneously. In automobile markets, engine capacity, power, physical dimensions, weight, fuel economy, drivetrain, body configuration, and brand-related characteristics may all be associated with price. A useful **pricing analysis** therefore requires more than identifying one high-correlation feature: it requires disciplined data preparation, exploration of competing relationships, explicit model specification, and evidence that fitted relationships generalize beyond the observations used to estimate them.

This project asks a focused empirical question: **which measurable automobile characteristics are most strongly associated with price, and which regression specification provides the strongest balance of explanatory usefulness, predictive generalization, and interpretability?** The objective is not to build a production-grade pricing engine. Instead, the study is designed as a transparent **regression analysis** that demonstrates the analytical reasoning required to move from raw data to defensible model selection.

The report follows a conventional empirical research structure - introduction, data and methods, results, discussion, limitations, and conclusion - while avoiding a cell-by-cell reproduction of the companion notebooks. This structure reflects common guidance for research and technical reports, in which methods establish reproducibility, results present the evidence, and discussion interprets its meaning and limitations.

1.1 Research Questions

- How is automobile price distributed in the analytical sample?
- Which continuous and categorical vehicle characteristics show the strongest empirical relationships with price?
- How much explanatory fit is obtained from simple linear, multiple linear, and polynomial regression specifications?
- Does additional polynomial flexibility improve generalization, or does it eventually produce overfitting?
- Which regression specification performs most consistently under cross-validation and on an untouched holdout sample?

2. Data Source and Analytical Sample

The project uses the **Automobile dataset** distributed by the **UCI Machine Learning Repository**. UCI describes the dataset as originating from the **1985 Ward's Automotive Yearbook**, with 205 instances, 25 features, mixed categorical/integer/real feature types, missing values, and regression among the associated analytical tasks. UCI states that the data combine automobile specifications, an insurance risk rating ("symboling"), and normalized loss information. The repository identifies Jeffrey Schlimmer as creator and provides DOI 10.24432/C5B01C under a CC BY 4.0 license.

The working CSV used in the project is the downloaded copy of the same automobile data and is read using an explicit 26-column schema that includes the price outcome. The first notebook verifies 205 rows and 26 columns, with no exact duplicate rows. Seven columns contain "?" missing-value markers, totaling 59 markers. The largest missingness occurs in normalized-losses (41 observations; 20.0%), while bore, stroke, and price each have four missing observations; horsepower, peak-rpm, and num-of-doors each have two.

Item	Value
Raw observations	205
Raw columns	26
Missing-value markers ("?")	59
Rows missing price	4

Cleaned observations	201
Cleaned columns	29
Exact duplicate rows	0

2.1 Data Preparation

Cleaning decisions were deliberately variable-specific. Question-mark markers were converted to standard missing values. Numerical predictors normalized-losses, bore, stroke, horsepower, and peak-rpm were mean-imputed to preserve observations in a small sample. The two missing num-of-doors values were replaced with the modal category. Price was treated differently: because it is the dependent variable for the regression analysis, the four observations without an observed price were removed rather than imputed. The resulting analysis-ready sample contains 201 observations and no remaining missing values.

Two interpretable feature-engineering steps were retained. City and highway fuel economy were converted to supplementary litres-per-100-kilometres variables while preserving the original MPG fields. Horsepower was also grouped into low, medium, and high bands for descriptive analysis, while the original continuous horsepower measure was retained for regression.

3. Methodology

The empirical workflow was designed to progress from **descriptive evidence** to **model specification** and then to **out-of-sample evaluation**. The notebooks use **Python** with **Pandas** and **NumPy** for data management, **Matplotlib** for visualization, **SciPy** for Pearson correlation testing, and **scikit-learn** for regression and validation.

3.1 Exploratory Analysis

EDA first characterizes the price distribution and then examines continuous predictors using **Pearson correlation coefficients** and **p-values**. **Scatterplots** are used to assess whether strong correlations correspond to systematic relationships or reveal nonlinearity and unusual observations. Categorical variables are summarized using counts, means, medians, standard deviations, boxplots, grouping, and pivot tables. A separate **predictor-correlation matrix** is used to identify overlap among candidate numerical predictors before multiple regression.

3.2 Regression Specifications

Four specifications are developed. The first **simple linear regression (SLR)** models price using **highway MPG**, preserving the original data's fuel-economy specification. A second SLR uses **engine size**, which EDA identifies as the strongest continuous bivariate correlate of price. The principal **multiple linear regression (MLR)** combines **horsepower**, **curb weight**, **engine size**, and **highway MPG**. A **polynomial highway-MPG model** is then used to test whether nonlinear curvature improves on the straight-line relationship.

3.3 Evaluation Strategy

Notebook 04 reports in-sample R^2 , MSE, RMSE, and MAE only as model-development diagnostics. Final model claims are based on **Notebook 05**. The cleaned data are split into 80% training and 20% holdout testing samples using `random_state = 42`. **Five-fold shuffled cross-validation** is conducted on the training sample only. The holdout **test set** remains untouched during model comparison and polynomial-degree selection.

Polynomial degrees 1 through 15 are compared by mean training R^2 , mean cross-validation R^2 , cross-validation R^2 variability, RMSE, and MAE. **Degree selection** is based on training-sample cross-validation rather than test-set performance. The selected polynomial is then refitted on the full training sample and evaluated once on the holdout test set. This design separates model development from final evaluation and makes overfitting visible.

4. Empirical Results

4.1 Price Distribution

The cleaned sample has a mean automobile price of \$13,207 and a median of \$10,295. Price skewness is 1.810, indicating a pronounced right tail. The gap between the mean and median confirms that a relatively small number of high-price vehicles pull the average upward. This distribution is important for later evaluation because squared-error metrics can be strongly influenced by expensive observations.

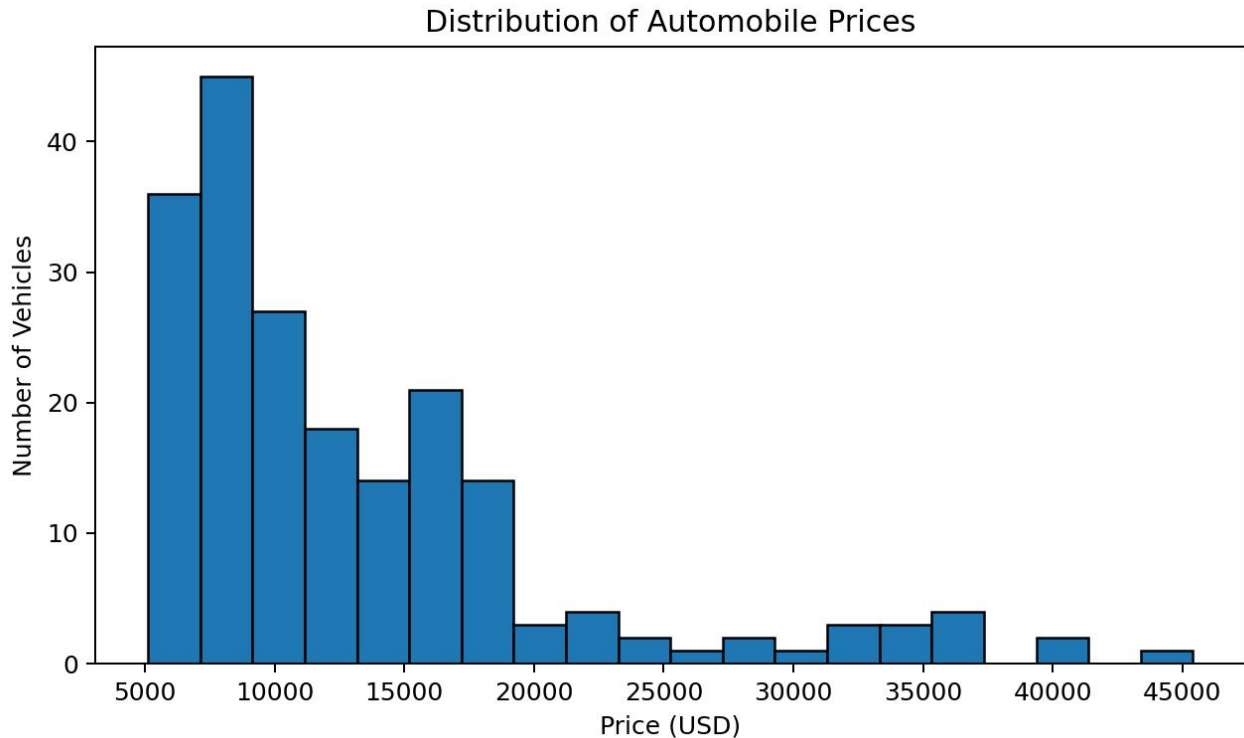


Figure 1. Distribution of automobile prices in the cleaned analytical sample.

4.2 Continuous Relationships with Price

The strongest continuous associations are concentrated in variables describing engine capacity, vehicle mass, power, and physical size. **Engine size** has the largest Pearson correlation with price ($r = 0.872$), followed by **curb weight** ($r = 0.834$), **horsepower** ($r = 0.810$), and **width** ($r = 0.751$). **Highway MPG** ($r = -0.705$) and **city MPG** ($r = -0.687$) are negatively associated with price. Length ($r = 0.691$), wheel base ($r = 0.585$), and bore ($r = 0.543$) also show statistically significant positive associations. Peak RPM, stroke, and compression ratio show weak linear correlations in this sample.

Feature	Pearson r	p-value
engine-size	0.872	<0.001
curb-weight	0.834	<0.001
horsepower	0.81	<0.001
width	0.751	<0.001
highway-mpg	-0.705	<0.001
length	0.691	<0.001
city-mpg	-0.687	<0.001

wheel-base	0.585	<0.001
bore	0.543	<0.001

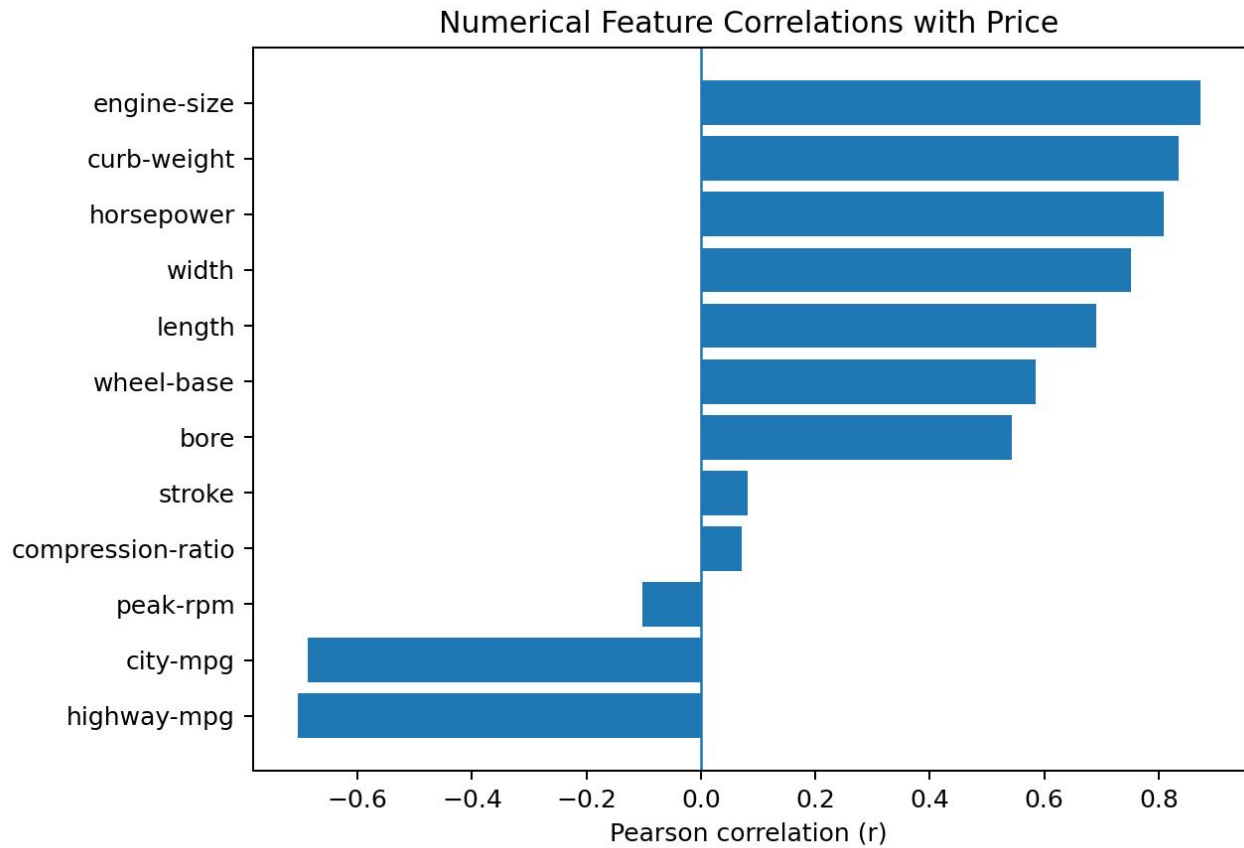


Figure 2. Pearson correlations between selected numerical characteristics and automobile price.

4.3 Categorical Price Differences

Categorical comparisons reinforce the view that price reflects product configuration as well as continuous technical characteristics. Rear-wheel-drive vehicles have a mean price of \$19,758 across 75 observations, compared with \$10,241 for the eight four-wheel-drive vehicles and \$9,245 for 118 front-wheel-drive vehicles. Body-style means are highest for hardtops (\$22,209) and convertibles (\$21,891), but these categories contain only eight and six observations respectively. Rear-engine vehicles have a much higher mean price (\$34,528), but only three rear-engine observations are present. These small groups are therefore descriptive signals rather than stable population estimates.

Variable	Category	n	Mean price
Drive wheels	rwd	75	\$19,758
Drive wheels	4wd	8	\$10,241
Drive wheels	fwd	118	\$9,245
Body style	hardtop	8	\$22,208
Body style	convertible	6	\$21,890
Engine location	rear	3	\$34,528
Engine location	front	198	\$12,884

4.4 Predictor Overlap and Modeling Implication

EDA also shows that several strong price predictors are correlated with one another. This is substantively plausible: larger engines often accompany higher horsepower and heavier vehicles, while city and highway MPG capture closely related fuel-economy characteristics. Consequently, a high bivariate correlation with price should not be interpreted as an independent contribution. This overlap motivates the transition from SLR to MLR and requires caution when interpreting individual MLR coefficients.

4.5 Regression Model Development

In-sample results show a clear progression in explanatory fit. **Highway MPG** alone explains approximately **49.7%** of observed price variation ($R^2 = 0.497$). **Engine size alone** performs substantially better ($R^2 = 0.761$). A **cubic polynomial** in **highway MPG** improves on the straight-line highway-MPG specification ($R^2 = 0.674$), but remains weaker than the engine-size SLR. The four-predictor MLR provides the strongest in-sample fit, with $R^2 = 0.809$, RMSE = \$3,461, and MAE = \$2,451.

Model	R2	RMSE	MAE
SLR - Highway MPG	0.497	\$5,625	\$4,055
SLR - Engine Size	0.761	\$3,876	\$2,795
MLR - 4 Predictors	0.809	\$3,461	\$2,451
Polynomial - Highway MPG (degree 3)	0.674	\$4,525	\$2,952

The highway-MPG SLR coefficient is -\$821.73 per additional MPG, consistent with the negative EDA relationship. The engine-size SLR coefficient is \$166.86 per one-unit increase in engine size. In the MLR, the fitted coefficients are 53.50 for horsepower, 4.71 for curb weight, 81.53 for engine size, and 36.06 for highway MPG, with an intercept of -\$15,806.63. These MLR coefficients are conditional associations and should not be ranked by raw magnitude because the predictors use different measurement scales and share information with one another.

4.6 Polynomial Complexity and Overfitting

Cross-validation provides a more informative view of polynomial complexity than in-sample fit alone. Validation performance improves from degree 1 (mean CV $R^2 = 0.444$) through **degree 6 (mean CV $R^2 = 0.657$; CV R^2 SD = 0.139)**. Degree 7 is nearly comparable, but validation R^2 then declines: 0.633 at degree 8, 0.576 at degree 9, and 0.440 at degree 10. At still higher degrees, validation performance becomes unstable and eventually strongly negative. The pattern provides direct evidence that additional polynomial flexibility eventually fits sample-specific structure rather than a stable highway-MPG/price relationship.

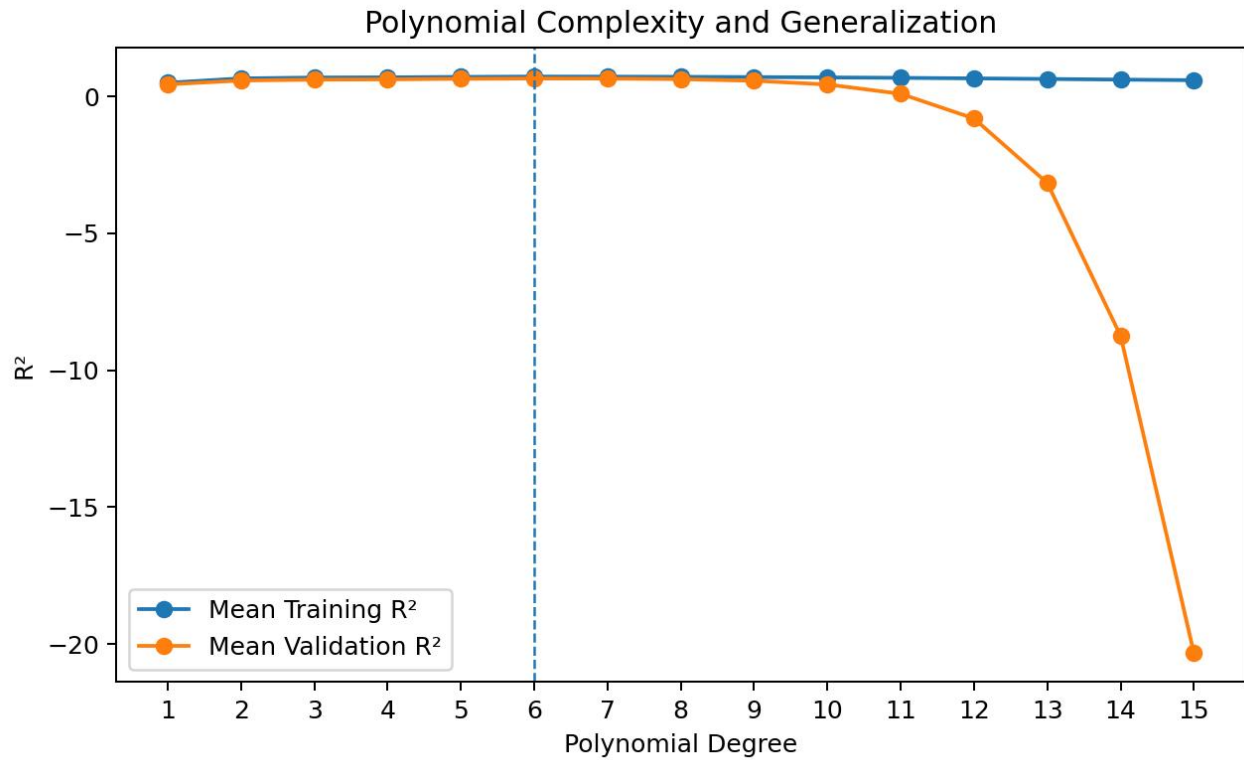


Figure 3. Training and cross-validation R^2 across polynomial degrees 1-15. Degree 6 has the highest mean validation R^2 .

Degree 6 is therefore selected from the training-sample cross-validation results. Its mean CV RMSE is \$3,669 and mean CV MAE is \$2,501. The fitted curve below is included in this report to make the selected nonlinear specification visually interpretable.

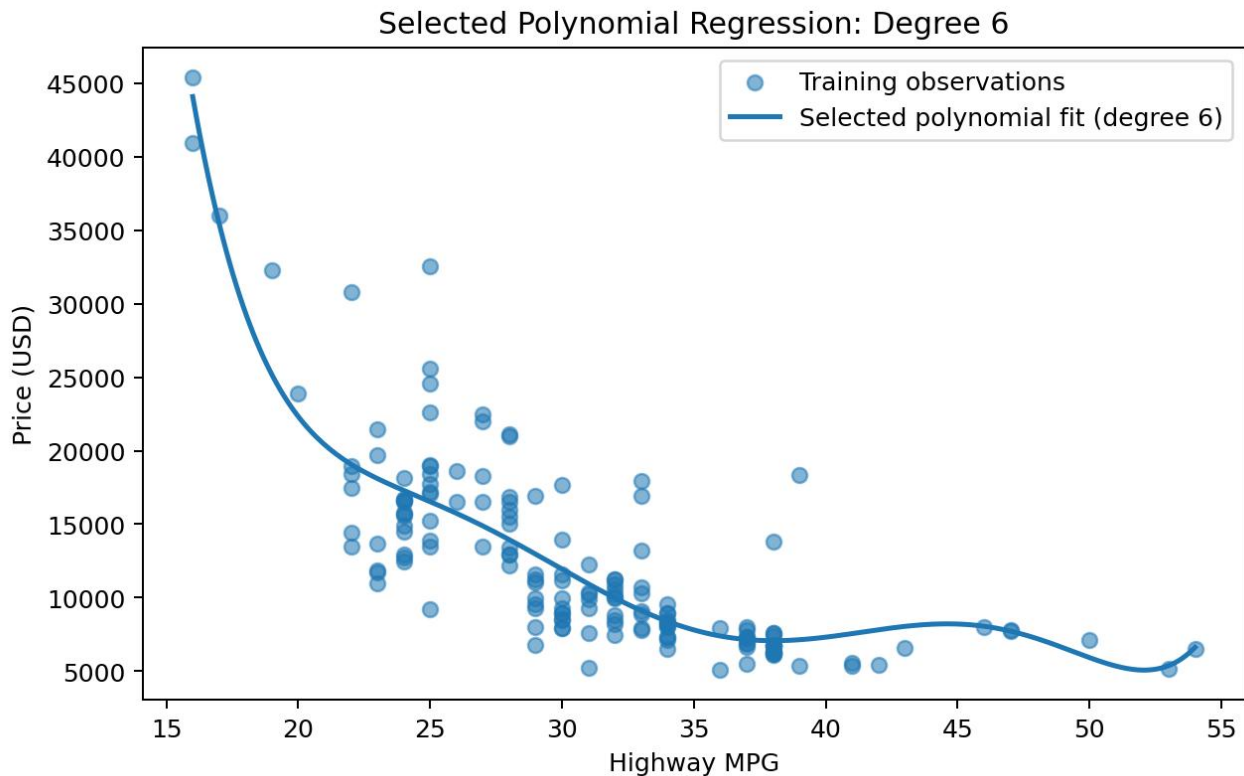
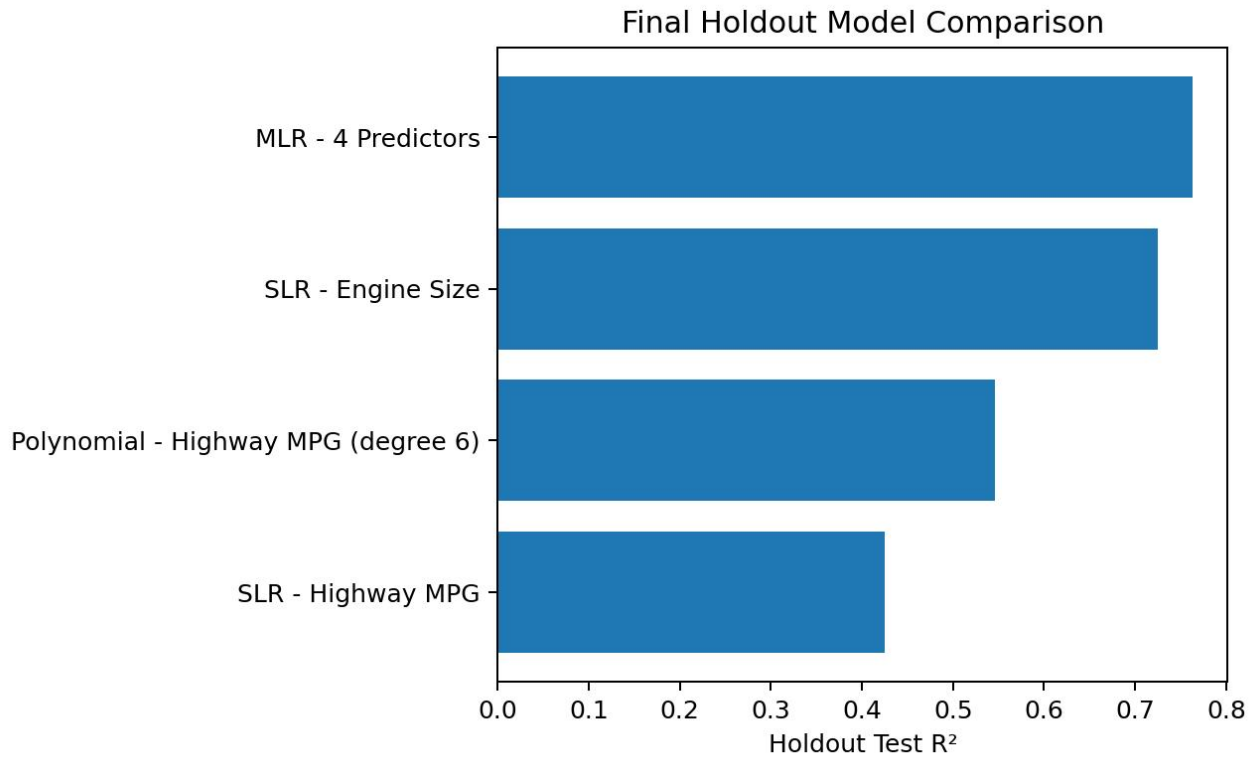


Figure 4. Selected degree-6 polynomial relationship between highway MPG and price, fitted on the training sample.

4.7 Final Out-of-Sample Comparison

The final holdout and cross-validation results identify the same leading specification. The **four-predictor MLR** achieves the highest **holdout R^2 (0.764)** and the highest **mean cross-validation R^2 (0.767)**. Its holdout RMSE is \$5,377 and MAE is \$3,769. The engine-size SLR ranks second on the holdout sample ($R^2 = 0.725$). Although the degree-6 polynomial improves materially over the linear highway-MPG model in cross-validation, its holdout R^2 is only 0.546, below both the MLR and engine-size SLR. The highway-MPG SLR is weakest (holdout $R^2 = 0.425$).

Model	Test R^2	Test RMSE	Test MAE	CV Mean R^2	CV R^2 SD	CV Mean RMSE	CV Mean MAE
MLR - 4 Predictors	0.764	\$5,377	\$3,769	0.767	0.078	\$3,092	\$2,197
SLR - Engine Size	0.725	\$5,805	\$4,176	0.701	0.149	\$3,381	\$2,582
Polynomial - Highway MPG (degree 6)	0.546	\$7,456	\$4,756	0.657	0.139	\$3,669	\$2,501
SLR - Highway MPG	0.425	\$8,390	\$5,963	0.444	0.110	\$4,846	\$3,383

Figure 5. Holdout test R^2 for the final candidate regression models.

5. Discussion

5.1 What the Results Suggest About Automobile Pricing

The evidence supports a multidimensional view of automobile price. Engine size is an unusually strong single predictor, but the best-performing specification combines engine size with horsepower, curb weight, and highway MPG. This result is analytically meaningful: price variation in the sample appears to reflect **multiple dimensions** of vehicle capacity, performance, physical mass, and efficiency rather than one isolated characteristic.

The negative bivariate relationship between MPG and price should be interpreted in context. In this historical sample, lower-MPG automobiles often coincide with larger, heavier, or more powerful vehicles. The positive highway-MPG coefficient in the MLR, despite its negative bivariate correlation with price, illustrates why multivariate interpretation differs from simple correlation. Once horsepower, curb weight, and engine size are held constant, the conditional coefficient can change sign because the predictors share substantial information. This is a useful demonstration of why business analysts should not equate pairwise correlation with an independent effect.

5.2 Why the MLR Is Preferred

The MLR is preferred not merely because it produces the highest score. It leads both the holdout and cross-validation comparisons, exhibits a relatively small training-to-test R^2 gap in the notebook evaluation, and remains straightforward to explain. The degree-6 polynomial demonstrates that nonlinear flexibility can improve a weak one-predictor model, but increasing complexity in highway MPG cannot substitute for information contained in several distinct vehicle characteristics. The comparison therefore favors broader explanatory information over increasingly complex curvature in a single feature.

5.3 Relevance to Business and Data Analytics

From a business analytics perspective, the project illustrates a generalizable analytical pattern. **Product pricing** questions often begin with heterogeneous raw data, incomplete observations, correlated attributes, and competing model specifications. The analyst's task is not only to fit a model but to establish data quality, identify interpretable empirical relationships, avoid leakage between model development and evaluation, assess generalization, and communicate limitations. Those capabilities are relevant to **pricing analytics**, **product strategy**, **market segmentation**, **risk analysis**, **demand modeling**, and other data-intensive management problems.

The project also demonstrates why **statistical modeling** and **predictive evaluation** should be integrated rather than treated as separate disciplines. EDA provides the substantive basis for feature selection; regression provides interpretable specifications; cross-validation tests whether apparent relationships are stable; and the holdout sample provides a final check against overconfident in-sample conclusions.

6. Limitations and Research Extensions

The analysis has several limitations that constrain interpretation and generalizability.

- Historical scope. The data originate from the 1985 Ward's Automotive Yearbook and should not be interpreted as a contemporary automobile market sample.
- Small sample. The cleaned analysis contains 201 observations, making estimates and holdout metrics sensitive to sample composition, particularly for rare categorical groups.
- Missing-data treatment. Mean imputation preserves observations but can reduce variance and attenuate natural relationships. This concern is most relevant to normalized-losses, which has 20% missingness.
- Observational design. The analysis identifies associations rather than causal effects. No research design is present that would justify causal interpretation of regression coefficients.

- Multicollinearity. Several vehicle characteristics are strongly related to one another. MLR coefficients therefore require cautious interpretation even when overall predictive performance is strong.
- Specification scope. The final regression models intentionally emphasize a compact set of continuous technical characteristics. Manufacturer and other categorical product attributes are not incorporated into the final MLR, so relevant price heterogeneity may remain unexplained.
- Error distribution and expensive vehicles. The right-skewed price distribution means high-price observations can have substantial influence on squared-error measures.

Natural extensions would include sensitivity analyses for alternative imputation strategies, explicit categorical controls, formal multicollinearity diagnostics, log-price specifications, robust regression diagnostics, and validation on a larger or more contemporary automobile dataset. Those extensions are intentionally left outside the current portfolio scope so that the project remains a focused demonstration of the core Python data-analysis and regression workflow.

7. Conclusion

This study develops a complete **Python-based regression analysis** from raw automobile data to final model selection. The workflow begins by auditing the source data, resolves missingness through explicit variable-specific rules, and creates an analysis-ready dataset. EDA then identifies strong price relationships involving engine size, curb weight, horsepower, vehicle width, and fuel economy while also revealing substantial overlap among predictors.

Regression development shows that **engine size** alone is a strong benchmark, but the **four-predictor MLR** provides the strongest overall representation of **price**. The final evaluation confirms that conclusion outside the estimation sample: the MLR achieves a holdout R^2 of 0.764 and mean 5-fold cross-validation R^2 of 0.767.

Polynomial analysis adds a complementary lesson. Increasing nonlinear complexity improves the highway-MPG model through degree 6, but higher degrees deteriorate sharply and the selected polynomial still underperforms the MLR on the holdout sample.

For a business/data analytics portfolio, the principal value of the project is the **analytical process** rather than the final coefficient table. It demonstrates **reproducible data importing, transparent cleaning, empirical exploration, regression specification, model diagnostics, cross-validation, overfitting analysis, holdout evaluation, and disciplined interpretation**. The analysis therefore serves as evidence of applied **Python** capability and of the statistical reasoning needed to connect quantitative methods to business research questions.

8. Reproducibility and Companion Notebooks

The report is a companion synthesis of five executed notebooks. Numerical claims in the report are based on their recorded outputs rather than recomputed or substituted external results. The notebook sequence is:

- 01 - Data Importing & Initial Understanding
- 02 - Data Cleaning & Feature Engineering
- 03 - Exploratory Data Analysis
- 04 - Regression Model Development
- 05 - Regression Model Evaluation & Selection

Together, the notebooks provide the reproducible analytical evidence; this report provides the integrated research narrative, interpretation, and portfolio-level communication.

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- Nikolov, P. Any empirical paper should roughly follow the format outlined below. Harvard University.
https://scholar.harvard.edu/files/pnikolov/files/empirical_paper_structure.pdf.
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<https://mitcommlab.mit.edu/broad/commkit/journal-article-results/>.

Appendix A. Key Analytical Results

The following values are reproduced from the executed companion notebooks and provide a compact audit trail for the report.

Result	Value
Raw data shape	205 rows x 26 columns
Cleaned data shape	201 rows x 29 columns
Mean price	\$13,207
Median price	\$10,295
Price skewness	1.810
Strongest price correlation	Engine size: $r = 0.872$
Notebook 04 best in-sample model	MLR: $R^2 = 0.809$
Selected polynomial degree	6
Selected polynomial mean CV R^2	0.6575
Final MLR mean CV R^2	0.767
Final MLR holdout R^2	0.764
Final MLR holdout RMSE	\$5,377
Final MLR holdout MAE	\$3,769